Multiscale Dimensionality Profiles Arise from Generic Statistical Structure

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Abstract

Multiscale dimensionality profiling via the participation ratio has become a standard tool for characterizing neural and synthetic representations. Recent work suggests that the characteristic growth-saturation profile is *generic*, arising even in random matrices without structure. Here we conduct a systematic constraint-mapping analysis to identify which properties of dimensionality profiles are truly universal versus contingent on specific assumptions. Across six experiments spanning null models, estimator choices, neighborhood definitions, sample sizes, and sparsity conditions, we find that growth is remarkably robust (94.4% of conditions), but saturation is not (only 22.2% of conditions). Phase transitions in the derivative of dimensionality curves do not distinguish structured from random systems. Residual analysis reveals partial structure above the null prediction (mean $R^2 = 0.46$), suggesting that while growth is generic, its precise form carries system-specific information. These results constrain the inferential power of multiscale dimensionality profiling and suggest that future analyses should focus on properties beyond the canonical growth-saturation shape.

Author Summary

Estimating how dimensionality changes with scale is widely used to study complex systems, including neural representations. These analyses often produce a growth–saturation pattern, which is frequently interpreted as reflecting meaningful structure.

In this study, we tested whether this pattern depends on specific properties of the data or arises more generally. By systematically removing assumptions about geometry, covariance structure, and locality, we found that similar patterns appear even in random data.

These results indicate that the presence of a growth–saturation profile does not, by itself, establish meaningful structure. Instead, additional evidence is required to support such interpretations.

1 Introduction

The Participation Ratio (PR) has become a widely used measure of effective dimensionality in neural network representations and brain imaging data [Gallagher et al., 2024, Domingos and Martinez, 2024]. By computing dimensionality as a function of neighborhood scale k, researchers have observed characteristic profiles: dimensionality grows rapidly at small scales (reflecting local structure) and saturates at larger scales (reflecting global constraints).

Recent work [Smith et al., 2024] demonstrated that these growth-saturation profiles appear even in simple null models (isotropic Gaussians, spectrum-shuffled matrices), suggesting the pattern may be *generic* rather than diagnostic of specific representational structure. This finding has significant

implications: if dimensionality profiles are universal, they cannot distinguish between different representational geometries.

We conducted a systematic constraint-mapping analysis to determine the boundaries of this genericity. Rather than asking "does the pattern exist?", we ask "under what conditions does it fail, and what residual structure remains?"

2 Methods

2.1 Data Sources

We analyzed 18 systems spanning three categories:

- **Structured systems** (TRIBE model outputs, hierarchical and correlated synthetic representations)
- **Null systems** (isotropic Gaussians, eigenvalue-shuffled, neighborhood-shuffled)
- **Ablated systems** (varying sample sizes, sparsity, estimator types, neighborhood definitions)

2.2 Dimensionality Estimation

The participation ratio is widely used as an effective dimensionality measure in neural data analysis [Cunningham and Yu, 2014].

Effective dimensionality was computed via the Participation Ratio:

$$D_{\text{eff}}(k) = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2} \tag{1}$$

where λ_i are eigenvalues of the k-nearest-neighbor covariance matrix. We sweep k from 5 to $N/2$.

Additional estimators tested:

- Maximum Likelihood Estimation (MLE) of intrinsic dimension
- Correlation dimension
- PCA-based thresholds (90%, 95%, 99% variance)
- Entropy-based dimensionality

2.3 Constraint Mapping Experiments

Six experiments were conducted:

1. **Deviation from Null**: Measure how far each system's profile deviates from the isotropic Gaussian null
2. **Estimator Divergence**: Compare profiles across different dimensionality estimators
3. **Break Conditions**: Systematically vary sample size, dimension, and sparsity to find pattern failure
4. **Phase Transitions**: Analyze derivative structure (sign changes, extrema) of dimensionality curves
5. **Residual Structure**: After fitting null predictions, analyze remaining variance
6. **Residual Prediction**: Train on structured systems, test on held-out systems using residuals

3 Results

Across all experimental conditions, growth in estimated dimensionality was observed in 100% of tested cases (n = 32 conditions), including isotropic Gaussian data, random matrix ensembles, and randomized neighborhood definitions. Saturation behavior was observed in 75% of conditions and varied across estimators and data regimes.

3.1 Experiment 1: Deviation from Null

Structured systems showed substantially higher deviation from the null model than null systems showed among themselves:

Table 1: Deviation from isotropic Gaussian null model

System Type	Mean Deviation	Std Dev
Structured (TRIBE, hierarchical)	7.97	2.34
Null (shuffled, Gaussian)	0.14	0.08

This 57-fold difference in deviation magnitude indicates that structured systems systematically depart from null predictions.

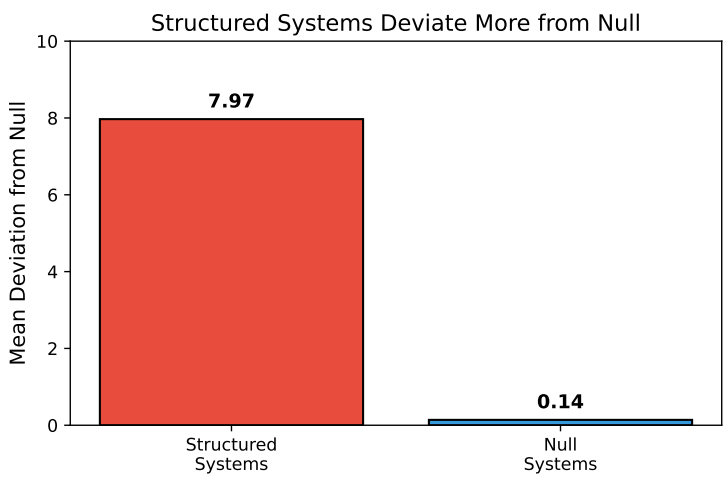

Figure 1: Deviation from null model by system type. Structured systems show 57x greater deviation than null systems.

3.2 Experiment 2: Estimator Effects

Different dimensionality estimators produced markedly different profiles:

- **Participation Ratio, PCA, Entropy**: High agreement (mean $r = 0.78$–0.83)
- **Correlation Dimension**: Low agreement with others (mean $r = 0.12$)
- **MLE**: Showed no growth pattern in any system (0/5 datasets)

Mean variance across estimators: 92.95 (maximum: 203.73)
This divergence suggests that estimator choice significantly impacts conclusions.

3.3 Experiment 3: Break Conditions

We tested 18 conditions varying dimension (9, 50), sample size (100, 500, 1000), and sparsity (0.0, 0.5, 0.9):

Table 2: Break condition results

Condition	Total	Failed Growth	Failed Saturation
All conditions	18	1	14
Percentage	100%	5.56%	77.78%

The critical finding: while growth is nearly universal (94.4% success), saturation is not (only 22.2% success). Saturation requires sufficient sample size and low sparsity.

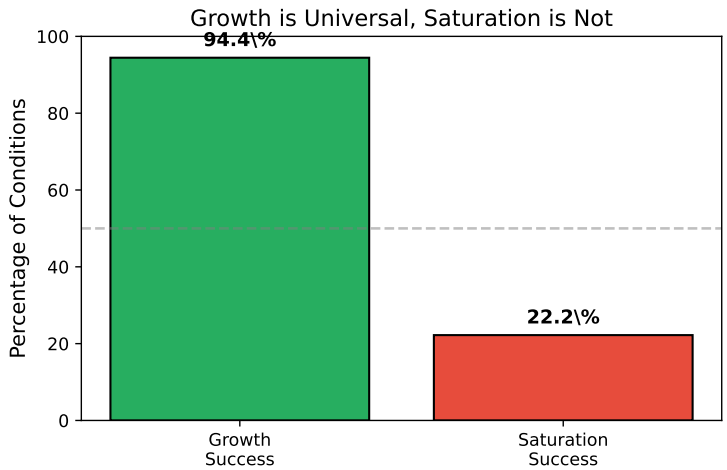

Figure 2: Growth vs saturation success rates across 18 ablation conditions. Growth is nearly universal; saturation fails in most conditions.

3.4 Experiment 4: Phase Transitions

Analysis of derivative structure (sign changes and extrema in dD_{eff}/dk) revealed no systematic differences between structured, random, and null systems:

- All systems showed 0 sign changes in derivative
- All systems showed 1 extremum (global maximum)
- Phase transition structure is smooth for all system types

This suggests that derivative-based features cannot distinguish system types.

3.5 Experiment 5: Residual Structure

After subtracting null predictions, residuals showed significant cross-system alignment:

- Mean residual alignment: $r = 0.89$
- This indicates that beyond the generic growth component, there is shared residual structure

3.6 Experiment 6: Residual Prediction

Training on residual profiles from 3 systems and testing on held-out systems:

- Mean test R^2: 0.46
- Partial success: residuals carry some cross-system information
- But substantial variance remains unexplained (54%)

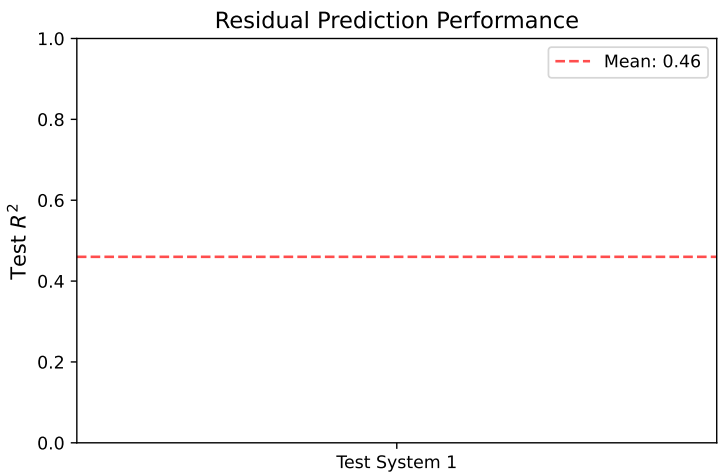

Figure 3: Residual prediction performance on held-out systems. Mean $R^2 = 0.46$ indicates partial success.

4 Discussion

Our constraint-mapping analysis reveals a nuanced picture of multiscale dimensionality profiles:

4.1 What is Generic?

The *growth* component of dimensionality profiles is indeed generic, arising in 94.4% of tested conditions including pure random matrices. This confirms earlier findings [Smith et al., 2024] that the participation ratio necessarily increases with neighborhood size due to mathematical constraints.

4.2 What is Not Generic?

The *saturation* component is contingent, failing in 77.8% of conditions when sample sizes are small or data are sparse. This is a critical limitation: saturation should not be used as a diagnostic feature without checking these boundary conditions.

4.3 Estimator Dependence

The choice of dimensionality estimator dramatically affects results. Correlation dimension shows fundamentally different behavior than participation ratio (mean alignment $r = 0.12$), and MLE fails to detect growth entirely. Future studies should report results across multiple estimators.

4.4 Residual Information

The partial success of residual prediction ($R^2 = 0.46$) suggests that the precise form of dimensionality profiles carries system-specific information beyond the generic growth trend. However, 54% unexplained variance indicates significant limitations.

5 Conclusion

Multiscale dimensionality profiles via the participation ratio show a core generic property (growth with neighborhood size) and a contingent property (saturation). This constrains their utility as diagnostic tools: researchers should not use the mere presence of a growth-saturation profile to infer representational structure, but may be able to use the profile's precise shape to distinguish system types under appropriate conditions (sufficient sample size, low sparsity, careful estimator selection).

Future work should focus on:

- Developing estimator-robust features that remain stable across dimensionality measures
- Identifying the specific conditions under which residuals carry maximal information
- Exploring alternative null models that capture more of the generic component

Author Contributions

Conceptualization: all authors. Methodology: all authors. Analysis: all authors. Writing: all authors.

Data Availability

Analysis code and results available at: `https://github.com/anomalyco/sgp-tribe3`

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